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Inventor(s)	Watariuchi; Satoki

Information processing apparatus, information processing system, and method for information processing apparatus

Abstract

A configuration is provided in which in a case where a user operation is performed on an image forming apparatus or a data processing apparatus while the image forming apparatus and the data processing apparatus are operating in cooperation with each other, the automatic clearing timers of both the apparatuses are reset.

Inventors:	**Watariuchi; Satoki (Tokyo, JP)**
Applicant:	**CANON KABUSHIKI KAISHA** (Tokyo, JP)
Family ID:	**83945081**
Assignee:	**Canon Kabushiki Kaisha (Tokyo, JP)**
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Primary Examiner: Augustin; Marcellus J

Attorney, Agent or Firm: Canon U.S.A., Inc. IP Division

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 17/936,768, filed Sep. 29, 2022, which claims the benefit of Japanese Patent Application No. 2021-161001, filed Sep. 30, 2021, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an information processing system in which apparatuses cooperate with each other to thereby execute a job.

Description of the Related Art

(2) With the recent widespread use of high-performance mobile terminals called smartphones, applications for using an image forming apparatus from such mobile terminals have been provided. For example, to execute a function of transmitting a scanned original document to a specified destination, an image forming apparatus connected to a local area network (LAN) is searched for, and a job to which the email address of the destination and information about various scan settings (for example, color and double-sided) set by the user are added is generated. The generated job is transmitted to the image forming apparatus that has been found and that has a transmission function to thereby allow the mobile terminal to instruct the image forming apparatus to perform a series of processes necessary for transmission. At this time, network communication between the mobile terminal and the image forming apparatus is established in accordance with a communication protocol, such as Transmission Control Protocol/Internet Protocol (TCP/IP), Near Field Communication (NFC), or Bluetooth (registered trademark), and can be implemented with a publicly known technique.

(3) A recent information processing apparatus, such as an image forming apparatus, has a function of clearing the set values of settings, restoring the display screen to an initial screen, or logging out in a case where no user operation on an operation unit is performed for a specific time. The function of restoring the apparatus to the initial state in a case where no operation is accepted for a specific time is defined as an automatic clearing function. Japanese Patent No. 3517183 proposes an image forming apparatus that can set the time (hereinafter referred to as "automatic clearing time") until the automatic clearing function is executed, on an operation mode by operation mode basis.

(4) In a case where an information processing apparatus is operated while the information processing apparatus and a terminal are made to cooperate with each other, there is a problem that automatic clearing not intended by the user may occur, resulting in decreased usability. For example, a case is assumed where a job is transmitted from the terminal to the information processing apparatus and various processes necessary for executing the job are performed by operating the terminal. The user operates an operation unit of the terminal but does not operate an operation unit of the information processing apparatus, and therefore, an automatic clearing process is performed in the information processing apparatus after a predetermined time elapsed.

SUMMARY OF THE INVENTION

(5) The present invention provides a method for allowing an automatic clearing process to be appropriately performed even in a case where an information processing apparatus and a terminal are made to cooperate with each other.

(6) The present invention provides an information processing apparatus configured to be capable of communicating with a terminal, the information processing apparatus including an operation unit, the information processing apparatus including: a counting unit configured to count a time during which no user operation on the operation unit is accepted; a performing unit configured to perform an automatic clearing process in a case where a predetermined time is counted by the counting unit; and a resetting unit configured to reset the time counted by the counting unit in response to a predetermined request received from the terminal.

(7) Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating a configuration of an information processing system.

(2) FIG. **2** is a block diagram illustrating a hardware configuration of a data processing apparatus.

(3) FIG. **3** is a block diagram illustrating a hardware configuration of an image forming apparatus.

(4) FIG. **4** is a block diagram illustrating a software configuration of the data processing apparatus.

(5) FIG. **5** is a block diagram illustrating a software configuration of the image forming apparatus.

(6) FIGS. **6**A and **6**B illustrate an example of a cooperation management table of the image forming apparatus in a first embodiment.

(7) FIG. **7** is a flowchart illustrating a process performed by the data processing apparatus in the first embodiment.

(8) FIGS. **8**A to **8**E illustrate examples of screens of the data processing apparatus in the first embodiment.

(9) FIGS. **9**A and **9**B are flowcharts illustrating processes performed by the data processing apparatus in the first embodiment.

(10) FIG. **10** is a flowchart illustrating a process performed by the data processing apparatus in the first embodiment.

(11) FIGS. **11**A and **11**B illustrate a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

(12) FIGS. **12**A to **12**E illustrate examples of screens of the image forming apparatus in the first embodiment.

(13) FIG. **13** is a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

(14) FIG. **14** is a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

(15) FIG. **15** is a flowchart illustrating a process performed by the data processing apparatus in a second embodiment.

(16) FIG. **16** is a flowchart illustrating a process performed by the image forming apparatus in the second embodiment.

(17) FIG. **17** is a sequence chart illustrating a process performed by the image forming apparatus and the data processing apparatus in the first embodiment.

DESCRIPTION OF THE EMBODIMENTS

(18) Hereinafter, embodiments of the present invention will be described with reference to the drawings. Note that configurations described in the embodiments are examples and are not intended to limit the scope of the present invention to the configurations. Although a description will be given below under the assumption that an image forming apparatus is an example of the information processing apparatus, the information processing apparatus is not limited to this.

First Embodiment

(19) FIG. **1** is a diagram illustrating a configuration of an information processing system according to this embodiment. An information processing system **100** includes a data processing apparatus **101**, a wireless LAN terminal **102**, and an image forming apparatus **104**, and these apparatuses are connected to each other over a network **103**, which is, for example, a LAN. To the network **103**, an apparatus other than those illustrated in FIG. **1** may be connected.

(20) The data processing apparatus **101** is, for example, a mobile terminal, such as a smartphone, or may be an information processing apparatus on which an operating system for small terminals and programs for controlling voice calls, position detection, and data communication run. Alternatively, the data processing apparatus **101** may be a personal computer (PC) that does not have functions of voice call control, position detection control, and so on. The data processing apparatus **101** is

connected to the network **103** via the wireless LAN terminal **102**.
(21) The wireless LAN terminal **102** is a wireless LAN base unit having a typical network router function and provides a wireless LAN, for example, at home or in an office.
(22) The image forming apparatus **104** is a multifunction peripheral (MFP) having various functions including a printer function, a copy function, a scanner function, and a facsimile transmission function. However, the image forming apparatus **104** is not limited to this and may be an apparatus having a single function. The image forming apparatus **104** is configured to include a wireless access point so as to be capable of directly communicating with the data processing apparatus **101**. A user can instruct the image forming apparatus **104** to execute jobs from the data processing apparatus **101** that communicates with the image forming apparatus **104** to thereby execute various functions of the image forming apparatus **104**. Although the image forming apparatus **104** is configured to be connected to the network **103** by wire in this embodiment, the image forming apparatus **104** may be connected to the network **103** via the wireless LAN terminal **102** similarly to the data processing apparatus **101**.
(23) Further, the data processing apparatus **101** and the image forming apparatus **104** can perform short-range wireless communication conforming to, for example, NFC or Bluetooth Low Energy (BLE). The image forming apparatus **104** transmits to the data processing apparatus **101** via an NFC communication unit or a BLE communication unit described below, connection information (an Internet Protocol (IP) address or a Media Access Control (MAC) address, a Service Set IDentifier (SSID), and so on) for a wireless LAN connection with the image forming apparatus **104**. The data processing apparatus **101** uses the obtained connection information to start wireless LAN communication with the image forming apparatus **104**.
(24) FIG. **2** is a diagram illustrating a hardware configuration of the data processing apparatus **101** according to this embodiment. The data processing apparatus **101** includes a controller unit **201**. The controller unit **201** controls various communication units including an NFC communication unit **210**, a BLE communication unit **211**, and a wireless LAN communication unit **212** and various user interface (UI) units including a microphone-speaker **213**, a display **214**, and a touch panel **215**.
(25) The controller unit **201** includes a central processing unit (CPU) **202**, a read-only memory (ROM) **203**, a random access memory (RAM) **204**, a network interface (UF) **205**, an audio control unit **206**, a display control unit **207**, an input control unit **208**, and a storage device **209**, and these are connected to each other via a system bus **217**.
(26) The CPU **202** controls the entire system of the data processing apparatus **101**. The ROM **203** stores the operating system of the data processing apparatus **101** and programs for controlling calls, data communication, and so on, and the CPU **202** executes the programs. The RAM **204** is used as a main memory of the CPU **202** and as a temporary storage area, such as a work area.
(27) The storage device **209** is a nonvolatile storage device to which settings in various operation modes, an operation log, and so on that need to be retained even after the restart of the data processing apparatus **101** are recorded.
(28) The network OF **205** is connected to the NFC communication unit **210**, the BLE communication unit **211**, and the wireless LAN communication unit **212** to exchange data with the image forming apparatus **104** through various types of wireless communication.
(29) The audio control unit **206** controls input and output of audio data via the microphone-speaker **213**. The display control unit **207** controls output of a screen to be displayed on the display **214**. The input control unit **208** controls input of information specified by the user using a button or the touch panel **215**. Various applications executed by the data processing apparatus **101** use the audio control unit **206**, the display control unit **207**, the input control unit **208**, and so on.
(30) FIG. **3** is a diagram illustrating a hardware configuration of the image forming apparatus **104** according to this embodiment. The image forming apparatus **104** includes a controller unit **301** and controls various communication units including an NFC communication unit **310**, a BLE communication unit **311**, and a wireless LAN communication unit **312**, an operation unit **307**, a

scanner **313** that executes the scan function, and a printer **314** that executes the print function. In a case where a user uses the copy function of the image forming apparatus **104**, the controller unit **301** controls the scanner **313** to obtain image data of an original document, and controls the printer **314** to print an image on a sheet and output the sheet. In a case where a user uses a "scan and transmit" function, the controller unit **301** controls the scanner **313** to obtain image data of an original document and convert the image data to code data. The controller unit **301** transmits the code data to, for example, the data processing apparatus **101** via a network I/F **308**. In a case where a user uses the print function, the controller unit **301** receives image data (code data) from the data processing apparatus **101** via the network I/F **308**. The controller unit **301** transmits the received image data to the printer **314**. The printer **314** prints an image on a sheet and outputs the sheet on the basis of the received image data. The image forming apparatus **104** further has a facsimile reception function of receiving data from, for example, an Integrated Services Digital Network (ISDN) and printing the data and a facsimile transmission function of transmitting scanned data to, for example, an ISDN. To execute each of the functions as described above, the image forming apparatus **104** executes a job, which is a processing target task for which the user performs a setting operation and so on, in accordance with job information. For example, in a case of a "scan and transmit" job, a scan is performed in accordance with scan settings included in job information, and image data generated as a result of the scan is transmitted to a destination included in the job information.

(31) The controller unit **301** includes a CPU **302**, a RAM **303**, a ROM **304**, a hard disk drive (HDD) **305**, an operation unit I/F **306**, the network I/F **308**, and a device I/F **309**, and these are connected to each other via a system bus **315**.

(32) The CPU **302** controls the entire system of the image forming apparatus **104**. The RAM **303** is a system work memory for the CPU **302** to operate, and temporarily stores image data and so on.

(33) The RAM **303** stores scan image data read by the scanner **313** and print image data received from the data processing apparatus **101** over the network **103**.

(34) The ROM **304** stores a boot program for the system, applications, and so on. The HDD **305** stores an operating system and system software, application software, image data, setting data, and so on.

(35) The operation unit **307** displays information for a user and accepts an operation from a user.

(36) The operation unit **307** includes an operation panel, such as a touch panel display, and hardware keys.

(37) The operation unit I/F **306** outputs to the operation unit **307** information to be displayed on the operation unit **307**.

(38) The operation unit I/F **306** accepts input by a user from the operation unit **307** information. The network I/F **308** is connected to the NFC communication unit **310**, the BLE communication unit **311**, and the wireless LAN communication unit **312** to exchange information with the data processing apparatus **101** or a cloud server through various types of communication. The wireless LAN communication unit **312** performs wireless LAN communication with the data processing apparatus **101** over the network **103**. The NFC communication unit **310** and the BLE communication unit **311** perform short-range wireless communication with the data processing apparatus **101**. The image forming apparatus **104** receives setting information and image data of a job, a job execution command, and so on from the data processing apparatus **101** via the network I/F **308** and executes the job. The device I/F **309** connects the scanner **313** and the printer **314** that respectively read and print image data with the controller unit **301** to input and output image data.

(39) FIG. **4** is a block diagram illustrating a software configuration of the data processing apparatus **101** according to this embodiment. FIG. **4** is a functional block diagram of software implemented by the CPU **202** reading a control program stored in the ROM **203** or the storage device **209**.

(40) An application **401** is installed in the data processing apparatus **101** and is an application for operating the image forming apparatus **104**. The application **401** communicates with the image

forming apparatus **104** and transmits various requests and receives various responses to and from the image forming apparatus **104** to thereby cooperate with the image forming apparatus **104**. For example, print job information or scan job information can be generated by performing a setting operation on the application **401**, and the job information can be transmitted to the image forming apparatus **104**. Although various applications are installed in the data processing apparatus **101** in addition to the application **401**, descriptions of such applications will be omitted. The configuration of the application **401** will be described below.

(41) A UI control unit **402** accepts setting information of a job and connection information of the image forming apparatus **104** input by the user on the touch panel **215** and transmits the information to an MFP control unit **403**. The UI control unit **402** receives a response from the MFP control unit **403** and outputs the response to the display **214** via the display control unit **207**. The UI control unit **402** further controls the automatic clearing function as part of user interface control. Automatic clearing is a process of clearing the values of all settings set on a display screen or displaying an initial screen set in advance in a case where the data processing apparatus **101** accepts no operation from the user for a predetermined time. The initial screen can be set in advance by the user, and in a case where the user remains logged in to the data processing apparatus **101** upon the occurrence of automatic clearing, a log-out process is automatically performed. A count-down timer for measuring the remaining time until the occurrence of automatic clearing is hereinafter referred to as "automatic clearing timer". The automatic clearing timer measures the time during which no operation is accepted from the user.

(42) The MFP control unit **403** establishes communication with the image forming apparatus **104**, and transmits a request and receives a response to and from the image forming apparatus **104**. The MFP control unit **403** generates job execution instruction information from job settings accepted by the UI control unit **402** and transmits a job execution request to the image forming apparatus **104** over the network **103**. The MFP control unit **403** obtains a job execution status from the image forming apparatus **104** over the network **103** and transmits the job execution status to the UI control unit **402**.

(43) FIG. **5** is a block diagram illustrating a software configuration of the image forming apparatus **104** according to this embodiment. System software **501** is software executed by the CPU **302** of the image forming apparatus **104** and is stored in the HDD **305**.

(44) A UI control unit **502** accepts information input by a user on the operation unit **307** and, for example, controls a screen. The UI control unit **502** controls the automatic clearing function as part of user interface control. In a case where a user remains logged in to the image forming apparatus **104** upon the occurrence of automatic clearing, the UI control unit **502** automatically performs a log-out process.

(45) A cooperation control unit **503** controls cooperation between the data processing apparatus **101** and the image forming apparatus **104**. Specifically, the cooperation control unit **503** controls the image forming apparatus **104** on the basis of a request received from the data processing apparatus **101** and transmits to the data processing apparatus **101** a response to the request. The cooperation control unit **503** manages the state (hereinafter referred to as "cooperation state") of the image forming apparatus **104** when the data processing apparatus **101** and the image forming apparatus **104** cooperate with each other.

(46) A cooperation management table **504** is a table for managing information necessary for the cooperation control unit **503** to control cooperation between the data processing apparatus **101** and the image forming apparatus **104**.

(47) A job control unit **505** controls processing units including a scan processing unit **511**, a copy processing unit **512**, a print processing unit **513**, and a facsimile processing unit **514** in accordance with job settings and executes various jobs.

(48) FIGS. **12**A to **12**E illustrate examples of screens displayed by the image forming apparatus **104** in this embodiment.

(49) FIG. **12**A is a diagram illustrating an authentication screen of the image forming apparatus **104** according to this embodiment. A log-in screen **1201** is a screen for a user authentication and includes a user name input field **1202** into which the user name of a user who locally logs in and a password input field **1203** into which a password is input. Local log-in is log-in for a user to exclusively operate the operation panel of the image forming apparatus **104**, and other users are not allowed to locally log in or operate the operation panel while the user remains locally logged in. When a log in button **1204** is pressed, the user name and the password respectively input into the user name input field **1202** and the password input field **1203** are used to start a local log-in process. Hereinafter, "log-in" refers to local log-in unless otherwise noted.
(50) FIG. **12**B is a diagram illustrating a home screen of the image forming apparatus **104**, and the home screen is displayed when, for example, authentication on the log-in screen **1201** is successful and the user logs in to the image forming apparatus **104**.
(51) A home screen **1210** is a screen for displaying functions of the image forming apparatus **104** in an array and is a screen for a user to select a function to be used. A copy button **1211** is a button for executing the copy function, and when pressed, a transition to a copy setting screen (not illustrated) takes place. When a check status button **1212** is pressed, a transition to a status check screen (not illustrated) takes place on which the user can refer to the execution status of a job and a history. A log out button **1213** displays the user name of a locally logged-in user, and when pressed, the logged-in user is made to log out and a transition to the log-in screen **1201** takes place. A "scan and transmit" button **1214** is a button for executing a function of transmitting a scanned image to a set destination. When the "scan and transmit" button **1214** is pressed, a transition to a "scan and transmit" setting screen **1220** takes place.
(52) FIG. **12**C is a diagram illustrating a screen of the "scan and transmit" function of the image forming apparatus **104**. The "scan and transmit" setting screen **1220** is a screen for setting a scan and a destination of an image. In a destination field **1221**, already set destinations of images are displayed. A specify destination button **1222** is a button for setting a destination of an image, and when pressed, a list of destinations is displayed. A scan setting area **1223** is an area for performing a setting operation for a scan, and settings can be specified by pressing respective buttons. A start button **1224** is a button for performing a process on the basis of the specifics set on the "scan and transmit" setting screen **1220**, and when pressed, a scan is performed and an image generated as a result of the scan is transmitted to a set destination. When the start button **1224** is pressed, the image forming apparatus **104** displays a process-in-progress screen (not illustrated) to show the status of the process that is being performed. When the process ends, the "scan and transmit" setting screen **1220** is displayed, or a transition to the home screen **1210** takes place. A stop button **1225** is a button for stopping the process that is in progress.
(53) FIG. **12**D illustrates an original-document placing screen for prompting a user to place an original document. An original-document placing screen **1231** is displayed in a case where an original document is not placed on the scanner of the image forming apparatus **104** upon reception of a "scan and transmit" job from the data processing apparatus **101**. When a cancel button **1232** is pressed, the job is canceled, and when an original document is placed or a placing completed button **1233** is pressed, the process starts.
(54) FIG. **12**E is a Personal Identification Number (PIN) code screen for displaying a PIN code to be input on the data processing apparatus **101**. This screen is displayed upon the image forming apparatus **104** receiving a "scan and transmit" job from the data processing apparatus **101**. To verify that the user of the data processing apparatus **101** is present in front of the image forming apparatus **104**, the user is made to input on the data processing apparatus **101** a PIN code displayed on a PIN code screen **1240**. The PIN code screen **1240** includes an area for displaying a PIN code generated in step S**1121** and a message for prompting the user to input the PIN code. A cancel button **1241** is a button for canceling the cooperation state. When a PIN code input on the data processing apparatus **101** is correct, the image forming apparatus **104** executes the "scan and

transmit" job.
(55) When the operation unit **307** accepts no operation from a user for a predetermined time while any of the screens illustrated in FIGS. **12**B to **12**E is displayed and an automatic clearing process is performed, the set values are cleared and the initial screen set in advance (for example, the home screen **1210**) is displayed. In a case where the user remains logged in to the image forming apparatus **104** upon the automatic clearing process, the user is made to log out and the log-in screen **1201** is displayed.
(56) FIGS. **8**A to **8**E illustrate examples of screens that are displayed in a case where the application **401** of the data processing apparatus **101** performs a process for instructing the image forming apparatus **104** to execute a job.
(57) FIG. **8**A illustrates an example of a job setting screen for "scan and transmit". A job setting screen **811** is displayed when a "scan and transmit" button **852** is selected from a menu screen **851** of the application **401**. The job setting screen **811** allows a setting operation for transmitting by email, an image generated by scanning an original document by the image forming apparatus **104**. In a title **812**, a title is displayed and a button for canceling settings and returning to the previous screen is provided. In an email address setting area **813**, already set destinations are displayed, and when this area is pressed, a software keyboard for inputting an email address of a destination is displayed. When an address book start button **814** is pressed, an address book in the data processing apparatus **101** is started, and the email address of a destination can be selected. A plurality of email addresses can be set. When an open button **815** is pressed, switching to a display screen that allows setting of CC, BCC, the subject, the body, and the file name takes place, and these items can be input. In a scan setting area **816**, buttons for specifying scan settings including a color setting and a resolution when the image forming apparatus **104** performs a scan are provided.
(58) When each of the buttons is pressed, a corresponding one of the settings can be changed. An execute job button **817** is a button for creating job information with the set values of the settings and transmitting the job information to the image forming apparatus **104**.
(59) FIG. **8**B illustrates an original-document placing screen that is a screen displayed in a case where an original document is not placed on the image forming apparatus **104** upon transmission of "scan and transmit" job information to the image forming apparatus **104**. A message area **821** is an area for displaying a message for prompting the user to place an original document. A cancel button **822** is a button for canceling the cooperation state. A placing completed button **823** is a button for notifying the image forming apparatus **104** that an original document has been placed, in a case where the user has placed the original document on the scanner **313** of the image forming apparatus **104** but the image forming apparatus **104** (scanner **313**) does not detect the original document.
(60) FIG. **8**C illustrates a PIN code input screen. This is a screen for inputting a PIN code included in the PIN code screen **1240** displayed on the image forming apparatus **104** after transmission of "scan and transmit" job information to the image forming apparatus **104**. A PIN code input field **831** is an input field for inputting a PIN code. A cancel button **832** is a button for canceling the cooperation state. An OK button **833** is a button for transmitting a PIN code input into the PIN code input field **831** to the image forming apparatus **104**. A configuration may be employed in which a QR code (registered trademark) is displayed on the image forming apparatus **104** instead of a PIN code and the QR code is read by a camera **216** included in the data processing apparatus **101** to thereby transmit authentication information.
(61) FIG. **8**D illustrates an end notification screen. This is a screen displayed when a PIN code input on the PIN code input screen is authenticated and job execution is started by the image forming apparatus **104**. A message area **841** is an area for displaying a message corresponding to a cooperation state received in step S**708**. In the message area **841** illustrated in FIG. **8**D, an example in a case where the cooperation state is "job execution" is displayed. An OK button **842** is a button for closing the end notification screen. When the end notification screen is closed, a transition to the menu screen of the application **401** takes place.

(62) FIG. **8**E illustrates the menu screen of the application **401**. The menu screen **851** is the initial screen that is displayed when the application **401** is started, and the user selects a desired function from the menu screen. When the “scan and transmit” button **852** is pressed, the job setting screen **811** is displayed.
(63) In a case where no operation is accepted from the user for a predetermined time while the data processing apparatus **101** displays any of the screens illustrated in FIGS. **8**A to **8**D, an automatic clearing process is performed, and the initial screen (for example, the menu screen illustrated in FIG. **8**E) is displayed.
(64) As described above, when a “scan and transmit” job is set on the application **401** and job information is transmitted to the image forming apparatus **104**, and an original document is placed and a PIN code is input, the job is executed by the image forming apparatus **104**.
(65) In a case where the data processing apparatus **101** is operated to make the image forming apparatus **104** perform a process, when the operation unit **307** of the image forming apparatus **104** accepts no operation from the user for a predetermined time and an automatic clearing process is performed as in the related art, a problem may arise. For example, when the image forming apparatus **104** determines that the operation unit **307** thereof accepts no operation even though the user performs an operation on the original-document placing screen illustrated in FIG. **8**B or the PIN code input screen illustrated in FIG. **8**C on the data processing apparatus **101**, keeps the automatic clearing timer counting, and performs an automatic clearing process, the original-document placing screen and the PIN code screen illustrated in FIG. **12**E are prevented from being displayed on the image forming apparatus **104** and, for example, the initial screen is displayed. Then, the user is, for example, unable to know a PIN code to be input and unable to make the process proceed. In this embodiment, when an operation is accepted by the data processing apparatus **101**, the image forming apparatus **104** is configured to reset the automatic clearing timer.
(66) FIG. **17** is a sequence chart illustrating interaction between the image forming apparatus **104** and the data processing apparatus **101** in this embodiment.
(67) When job settings from the user are accepted on the application **401** of the data processing apparatus **101** and an instruction for transmitting job information to the image forming apparatus **104** is accepted, the data processing apparatus **101** resets the automatic clearing timer thereof in S**1701**, and a job execution request is transmitted from the data processing apparatus **101** to the image forming apparatus **104** in S**1702**. That is, job information for making the image forming apparatus **104** execute a job is transmitted to the image forming apparatus **104**. The remaining time of the automatic clearing timer decreases as time passes, and when the remaining time becomes equal to zero, automatic clearing occurs. When the automatic clearing timer is reset, the remaining time until the occurrence of automatic clearing is reset to the automatic clearing time. For example, in a case where the automatic clearing time is set to 120 seconds, the remaining time is reset to 120 seconds when the automatic clearing timer is reset.
(68) When receiving the job execution request, the image forming apparatus **104** resets the automatic clearing timer thereof in S**1703**. In a case where an original document is placed on the image forming apparatus **104**, the cooperation state is changed to “PIN code request” in S**1704**, and a response is transmitted to the data processing apparatus **101** in S**1705**. The cooperation state will be described below.
(69) The data processing apparatus **101** transmits a cooperation state obtaining request to the image forming apparatus **104** in S**1706**, and the image forming apparatus **104** returns a response in S**1707**. In S**1708**, the data processing apparatus **101** determines whether the cooperation state has changed on the basis of the received response information, and in a case where the cooperation state has changed, the data processing apparatus **101** resets the automatic clearing timer in S**1709** and performs display corresponding to the changed cooperation state. It is assumed that the request and response process in S**1706** and S**1707** is performed occasionally after transmission of the job information.

(70) It is assumed that the process in S**1710** and the subsequent steps is performed asynchronously to the steps described above.
(71) When pressing of the placing completed button **823** by the user is accepted on the application **401** of the data processing apparatus **101**, the data processing apparatus **101** resets the automatic clearing timer thereof in S**1710** and transmits an original-document placing request to the image forming apparatus **104** in S**1711**.
(72) When receiving the original-document placing request, the image forming apparatus **104** resets the automatic clearing timer thereof in S**1712**. The image forming apparatus **104** changes the cooperation state to “PIN code request” in S**1713** and transmits an original-document placing response to the data processing apparatus **101** in S**1714**.
(73) When accepting a PIN code input by the user on the application **401** of the data processing apparatus **101**, the data processing apparatus **101** resets the automatic clearing timer thereof in S**1715** and transmits an unlock-with-PIN request to the image forming apparatus **104** in S**1716**.
(74) When receiving the unlock-with-PIN request, the image forming apparatus **104** resets the automatic clearing timer thereof in S**1717**. The image forming apparatus **104** changes the cooperation state to “job execution” in S**1718**, transmits an unlock-with-PIN response to the data processing apparatus **101** in S**1719**, and executes the job in S**1720** on the basis of the received job information.
(75) FIGS. **6**A and **6**B illustrate an example of the cooperation management table **504** in this embodiment. The cooperation management table **504** includes two tables, namely a request management table **600** and an event management table **650**.
(76) FIG. **6**A illustrates an example of the request management table **600** for managing requests received by the image forming apparatus **104** from the data processing apparatus **101**. One row defines information about one type of request.
(77) A “request type” column **601** includes the type of a request received by the image forming apparatus **104**.
(78) A “job execution request” is a request for instructing the image forming apparatus **104** to execute a job. A “cooperation state obtaining request” is a request for obtaining state information from the image forming apparatus **104**. An “original-document placing request” is a request for a notification that an original document has been placed. An “unlock-with-PIN request” is a request for performing authentication with an input PIN code. A “cancel request” is a request for canceling execution of a job.
(79) A “cooperation state for acceptance” column **602** defines a cooperation state in which the request included in the “request type” column **601**, when received, can be accepted. The table shows, for example, that the “job execution request” can be accepted only in a case where the cooperation state is “(not in cooperation)”. The cooperation state “(not in cooperation)” refers to a state in which, for example, the data processing apparatus **101** and the image forming apparatus **104** communicate with each other but job information is not transmitted from the data processing apparatus **101** to the image forming apparatus **104**, and a cooperation state “(in cooperation)” refers to a state in which job information has been transmitted from the data processing apparatus **101** to the image forming apparatus **104** and the apparatuses can cooperate with each other. As described above, the cooperation state indicates a stage or a status in a process performed by exchanging information between the data processing apparatus **101** and the image forming apparatus **104**. The “cooperation state obtaining request” can be accepted only in a case where a job has been transmitted from the data processing apparatus **101** to the image forming apparatus **104** and the cooperation state is “(in cooperation)”.
(80) A “user-operation-causing request” column **603** defines whether the request in the “request type” column **601** is a request caused by a user operation. In a case of a row in which the “user-operation-causing request” column **603** includes “YES”, the request is a request caused by a user operation. In a case of a row in which the “user-operation-causing request” column **603** includes

"NO", the request is a request transmitted regardless of a user operation. For example, the "job execution request" is a request that is transmitted when the user presses a button for transmitting a job on the data processing apparatus **101**, and therefore, "YES" is set in the "user-operation-causing request" column **603**. The "cooperation state obtaining request" is not a request that is transmitted when the user operates the data processing apparatus **101**, and therefore, "NO" is set in the "user-operation-causing request" column **603**.
(81) FIG. **6**B illustrates an example of the event management table **650** for managing event information about an event based on which the cooperation state changes. One row defines information about one cooperation state, event information about an event that is handled in the cooperation state, and a changed cooperation state after the occurrence of the event. A "cooperation state" column **651** includes one cooperation state. An "occurring event" column **652** defines an event to be handled when the image forming apparatus **104** is in the state included in the "cooperation state" column **651**, and the cooperation state is not changed in a case of the occurrence of an event not defined in the "occurring event" column **652**.
(82) A "changed cooperation state" column **653** defines a cooperation state after the occurrence of the event defined in the "occurring event" column **652**. A description of, for example, a row **662** will be given. In a case where either a "scanner: an original document placed" event or a "UI: the placing completed button pressed" event occurs while the cooperation state is "waiting for an original document to be placed", the cooperation state transitions to "PIN code request".
(83) With reference to FIG. **7** and FIGS. **8**A to **8**E, a process for a job execution instruction performed by the data processing apparatus **101** of this embodiment will be described. FIG. **7** is a flowchart for explaining the process for a job execution instruction, and the process is implemented by the CPU **202** of the data processing apparatus **101** loading a program for implementing a control module stored in the ROM **203** or the storage device **209** to the RAM **204** and executing the program.
(84) In step S**701**, the UI control unit **402** accepts information (an IP address and so on) for a connection with the image forming apparatus **104** that executes a job, the information being input by the user performing an operation on a screen of the application **401**. The information for a connection with the image forming apparatus **104** need not be input by the user but may be obtained through short-range wireless communication, such as NFC. The wireless LAN communication unit **212** may be used to search for an image forming apparatus that is connected to the network **103**. It is assumed that the information accepted in step S**701** is used to communicate with the image forming apparatus **104** in the subsequent steps.
(85) In step S**702**, the UI control unit **402** accepts job settings and a transmission instruction for job information input by the user performing an operation on the job setting screen **811**. Specifically, scan settings and a destination setting are accepted on the job setting screen **811**, and pressing of the execute job button **817** is accepted.
(86) In step S**703**, the UI control unit **402** resets the automatic clearing timer of the data processing apparatus **101**.
(87) In step S**704**, the MFP control unit **403** transmits a job execution request that includes the job information created on the basis of the operation accepted in step S**702** to the image forming apparatus **104**.
(88) In step S**705**, the MFP control unit **403** receives a job execution response from the image forming apparatus **104** as a response to the request transmitted in step S**704**. In step S**706**, the MFP control unit **403** determines the content of the job execution response. If the content of the job execution response indicates a success, the MFP control unit **403** makes the process proceed to step S**707**; otherwise the MFP control unit **403** makes the process proceed to step S**750**.
(89) In step S**707**, the MFP control unit **403** transmits a cooperation state obtaining request for obtaining the cooperation state to the image forming apparatus **104**. In step S**708**, the MFP control unit **403** receives a cooperation state obtaining response from the image forming apparatus **104** as a

response to the request transmitted in step S**707**. The cooperation state obtaining response includes the cooperation state of the image forming apparatus **104** and other information.
(90) In step S**709**, the MFP control unit **403** determines whether the cooperation state received in step S**708** has changed from the immediately preceding cooperation state. If the cooperation state has changed, the UI control unit **402** makes the process proceed to step S**710**; otherwise the UI control unit **402** makes the process return to step S**707**.
(91) In step S**710**, the UI control unit **402** resets the automatic clearing timer of the data processing apparatus **101** in response to the changed cooperation state. Although a change in the cooperation state is used as an example of the basis for determining whether to reset the automatic clearing timer upon reception of the cooperation state obtaining response in this flowchart, another basis for determination may be used. For example, determination may be performed on the basis of the type of the cooperation state included in the cooperation state obtaining response or determination may be performed by using other information (for example, information about whether a user operation is performed on the image forming apparatus **104**) included in the cooperation state obtaining response.
(92) In step S**711**, the MFP control unit **403** makes the process proceed to step S**712** if the cooperation state received in step S**708** is "waiting for an original document to be placed", makes the process proceed to step S**720** if the cooperation state received in step S**708** is "PIN code request", or makes the process proceed to step S**730** if the cooperation state received in step S**708** is "job execution" or "cancel".
(93) In step S**712**, the UI control unit **402** displays the original-document placing screen (FIG. **8**B) on the display **214**, and subsequently, makes the process return to step S**707**. The original-document placing screen is a screen for displaying a message for prompting the user to place an original document on the scanner **313** of the image forming apparatus **104**. A process that is performed in a case where a user operation is performed on the original-document placing screen will be described below with reference to the flowchart in FIG. **9**A.
(94) In step S**720**, the UI control unit **402** displays the PIN code input screen (FIG. **8**C) on the display **214**, and subsequently, makes the process return to step S**707**. The PIN code input screen is a screen for inputting a PIN code displayed on the operation unit **307** of the image forming apparatus **104**. A process that is performed in a case where a user operation is performed on the PIN code input screen will be described below with reference to the flowchart in FIG. **9**B.
(95) In step S**730**, the UI control unit **402** displays the end notification screen (FIG. **8**D) on the display **214** and ends the process in the flowchart. The end notification screen is a screen for notifying the user of the end state when the cooperation state with the image forming apparatus **104** ends.
(96) In step S**750**, the UI control unit **402** displays on the display **214** a message saying that the job execution instruction results in an error and ends the process in the flowchart.
(97) With the above-described process in the flowchart, even if the operation unit of the data processing apparatus **101** accepts no operation from the user for a predetermined time during cooperation between the data processing apparatus **101** and the image forming apparatus **104**, an automatic clearing process can be prevented from being performed in the data processing apparatus **101**.
(98) With reference to FIGS. **9**A and **9**B, processes performed in a case where user operations are performed on the application **401** will be described. FIGS. **9**A and **9**B are flowcharts for explaining processes performed in the data processing apparatus **101**, and the processes are implemented by the CPU **202** of the data processing apparatus **101** loading a program for implementing a control module stored in the ROM **203** or the storage device **209** to the RAM **204** and executing the program. The processes in the flowcharts illustrated in FIGS. **9**A and **9**B are performed asynchronously to the process in the flowchart illustrated in FIG. **7**.
(99) FIG. **9**A is a flowchart for explaining a process performed in a case where a user operation is

performed on the original-document placing screen (FIG. **8**B) displayed in step S**712**.
(100) In step S**901**, the UI control unit **402** accepts a user operation performed on the original-document placing screen and the process proceeds to the next step.
(101) In step S**902**, the UI control unit **402** resets the automatic clearing timer of the data processing apparatus **101**. In step S**903**, the UI control unit **402** determines the user operation detected in step S**901**. The UI control unit **402** makes the process proceed to step S**904** if the user operation is pressing of the placing completed button **823**, or makes the process proceed to step S**910** if the user operation is pressing of the cancel button **822**.
(102) In step S**904**, the MFP control unit **403** transmits to the image forming apparatus **104** an original-document placing request for giving a notification that an original document has been placed. In step S**905**, the MFP control unit **403** receives from the image forming apparatus **104** an original-document placing response as a response to the request transmitted in step S**904** and ends the process in the flowchart.
(103) In step S**910**, the MFP control unit **403** transmits to the image forming apparatus **104** a cancel request for canceling the cooperation state. In step S**911**, the MFP control unit **403** receives a cancel response from the image forming apparatus **104** as a response to the request transmitted in step S**910** and ends the process in the flowchart.
(104) FIG. **9**B is a flowchart for explaining a process performed in a case where a user operation is performed on the PIN code input screen (FIG. **8**C) displayed in step S**720**.
(105) In step S**920**, the UI control unit **402** accepts a user operation performed on the PIN code input screen, and the process proceeds to the next step.
(106) In step S**921**, the UI control unit **402** resets the automatic clearing timer of the data processing apparatus **101**. In step S**922**, the UI control unit **402** determines the user operation accepted in step S**920**. The UI control unit **402** makes the process return to step S**920** if the user operation is input of a PIN code, makes the process proceed to step S**923** if the user operation is pressing of the OK button **833**, or makes the process proceed to step S**930** if the user operation is pressing of the cancel button **832**.
(107) In step S**923**, the MFP control unit **403** transmits to the image forming apparatus **104** an unlock-with-PIN request for unlocking with the PIN code. The unlock-with-PIN request includes the PIN code input into the PIN code input field **831** by the user. In step S**924**, the MFP control unit **403** receives an unlock-with-PIN response from the image forming apparatus **104** as a response to the request transmitted in step S**923**. The unlock-with-PIN response includes information about whether unlocking with the PIN code is successful. In step S**925**, the MFP control unit **403** ends the process in the flowchart if unlocking with the PIN code is successful or makes the process proceed to step S**926** if unlocking with the PIN code fails. In step S**926**, the UI control unit **402** displays a message saying that unlocking with the PIN code fails and prompting the user to re-input a PIN code into a message area (PIN code input field **831**), and ends the process in the flowchart.
(108) In step S**930**, the MFP control unit **403** transmits to the image forming apparatus **104** a cancel request for canceling the cooperation state. In step S**931**, the MFP control unit **403** receives a cancel response from the image forming apparatus **104** as a response to the request transmitted in step S**930** and ends the process in the flowchart.
(109) With reference to FIG. **10**, a process for automatic clearing control in the data processing apparatus **101** will be described. FIG. **10** is a flowchart for explaining a process performed by the data processing apparatus **101**, and the process is implemented by the CPU **202** of the data processing apparatus **101** loading a program for implementing a control module stored in the ROM **203** or the storage device **209** to the RAM **204** and executing the program. The process in the flowchart illustrated in FIG. **10** is performed asynchronously to the process in the flowchart illustrated in FIG. **7** and to the processes in the flowcharts illustrated in FIGS. **9**A and **9**B.
(110) Note that in this embodiment, an example case where the menu screen of the application **401** is displayed in response to the occurrence of automatic clearing in the data processing apparatus

101 will be described. However, this embodiment is not limited to this, and the home screen of the data processing apparatus **101** may be displayed or the data processing apparatus **101** may be locked in a case where the data processing apparatus **101** is a mobile terminal.

(111) In step S**1001**, the UI control unit **402** detects the application **401** being called and displayed on the display **214**. In step S**1002**, the UI control unit **402** starts counting down by the automatic clearing timer. Thereafter, the automatic clearing timer decreases the remaining time as time passes.

(112) In step S**1003**, the UI control unit **402** monitors the remaining time of the automatic clearing timer, and makes the process proceed to step S**1004** when detecting the remaining time being equal to zero or keeps monitoring the remaining time if the remaining time is more than zero.

(113) In step S**1004**, the UI control unit **402** displays the menu screen **851** of the data processing apparatus **101** and ends the process in the flowchart.

(114) With the process in the flowchart described above, an automatic clearing process is performed when a predetermined time elapses in a state in which a predetermined condition is satisfied. Although an example of counting down has been described here, counting up may be performed and automatic clearing may be made to occur when the count reaches a predetermined count.

(115) With reference to FIGS. **11**A and **11**B and FIG. **12**, a cooperation process that is performed by the image forming apparatus **104** of this embodiment with the data processing apparatus **101** will be described. FIGS. **11**A and **11**B illustrate a flowchart for explaining the cooperation process in the image forming apparatus **104**, and the process is implemented by the CPU **302** of the image forming apparatus **104** loading a program for implementing a control module stored in the ROM **304** or the HDD **305** to the RAM **303** and executing the program.

(116) In step S**1101**, the cooperation control unit **503** receives over the network **103** a request transmitted from the data processing apparatus **101**. This request is, for example, information transmitted in step S**704** or step S**707**.

(117) In step S**1102**, the cooperation control unit **503** obtains from the request management table **600** information (information defined in the “cooperation state for acceptance” column **602** and in the “user-operation-causing request” column **603**) corresponding to the request received in step S**1101**.

(118) In step S**1103**, the cooperation control unit **503** obtains the current cooperation state of the image forming apparatus **104**. In step S**1104**, the cooperation control unit **503** determines whether the request received in step S**1101** is a request that can be accepted in the current cooperation state. Specifically, the cooperation control unit **503** determines whether the current cooperation state satisfies the value in the “cooperation state for acceptance” column **602** obtained in step S**1102**. If the request is a request that can be accepted, the cooperation control unit **503** makes the process proceed to step S**1105**; otherwise the cooperation control unit **503** makes the process proceed to step S**1180**.

(119) In step S**1105**, the cooperation control unit **503** determines whether the request received in step S**1101** is a request made by a user operation. Specifically, the cooperation control unit **503** performs determination on the basis of whether the value in the “user-operation-causing request” column **603** obtained in step S**1102** is “YES” or “NO”. Performing determination based on the value in the “user-operation-causing request” column **603** as to whether the request is a request made by a user operation is an example, and other methods may be used. For example, a request transmitted by the data processing apparatus **101** may include information indicating whether the request is a request made by a user operation, and determination may be performed on the basis of the information. If the request is a request made by a user operation, the cooperation control unit **503** makes the process proceed to step S**1106**; otherwise the cooperation control unit **503** makes the process proceed to step S**1107**.

(120) In step S**1106**, the UI control unit **502** resets the automatic clearing timer of the image forming apparatus **104**.

(121) In step S**1107**, the cooperation control unit **503** makes the process diverge in accordance with the type of the request received in step S**1101**. If the request is a “job execution request”, the cooperation control unit **503** makes the process proceed to step S**1108**. If the request is an “original-document placing request”, the cooperation control unit **503** makes the process proceed to step S**1120**. If the request is a “cancel request”, the cooperation control unit **503** makes the process proceed to step S**1130**. If the request is an “unlock-with-PIN request”, the cooperation control unit **503** makes the process proceed to step S**1140**. If the request is a “cooperation state obtaining request”, the cooperation control unit **503** makes the process proceed to step S**1170**.
(122) In step S**1108**, the cooperation control unit **503** saves job information included in the job execution request received in step S**1101** in the RAM **303**. The cooperation control unit **503** retains the saved job information until the cooperation state is canceled. In step S**1109**, the cooperation control unit **503** determines the original-document detection state of the scanner **313** of the image forming apparatus **104**. If the original-document detection state of the scanner **313** is a state in which a placed original document is detected, the cooperation control unit **503** makes the process proceed to step S**1120**; otherwise the cooperation control unit **503** makes the process proceed to step S**1110**.
(123) In step S**1110**, the cooperation control unit **503** changes the cooperation state to the “waiting for an original document to be placed” state.
(124) In step S**1111**, the UI control unit **502** displays the original-document placing screen (FIG. **12**D) on the operation unit **307** of the image forming apparatus **104**, and subsequently, makes the process proceed to step S**1150**. The original-document placing screen is a screen for displaying a message for prompting the user to place an original document on the scanner **313**. A process that is performed in a case where a user operation is performed on the original-document placing screen will be described with reference to the flowchart in FIG. **13**.
(125) In step S**1120**, the cooperation control unit **503** changes the cooperation state to the “PIN code request” state. In step S**1121**, the cooperation control unit **503** generates a PIN code. Although the PIN code may be generated in any manner, in this embodiment, for example, a four-digit number generated at random may be used as the PIN code. In step S**1122**, the UI control unit **502** displays the PIN code screen (FIG. **12**E) on the operation unit **307** of the image forming apparatus **104**, and subsequently, makes the process proceed to step S**1150**. The PIN code screen is a screen for presenting to the user the PIN code generated in step S**1121**. A process that is performed in a case where a user operation is performed on the PIN code screen will be described with reference to the flowchart in FIG. **13**.
(126) In step S**1130**, the cooperation control unit **503** changes the cooperation state to the “cancel” state. In step S**1131**, the cooperation control unit **503** sets a timer for canceling the cooperation state, and subsequently, makes the process proceed to step S**1150**. After the elapse of the time set for the timer, the cooperation control unit **503** deletes the cooperation state and the job settings saved in step S**1108**, and the image forming apparatus **104** transitions to the “not in cooperation” state.
(127) In step S**1140**, the cooperation control unit **503** determines whether a PIN code included in the unlock-with-PIN request received in step S**1101** matches with the PIN code generated in step S**1121**. If the PIN codes match, the cooperation control unit **503** makes the process proceed to step S**1141**; otherwise the cooperation control unit **503** makes the process proceed to step S**1160**.
(128) In step S**1141**, the cooperation control unit **503** changes the cooperation state to the “job execution” state. In step S**1142**, the cooperation control unit **503** requests the job control unit **505** to start executing the job by using the job settings saved in step S**1108**. Subsequently, the process proceeds to step S**1150**.
(129) In step S**1150**, the cooperation control unit **503** transmits to the data processing apparatus **101** a response to the request received in step S**1101** and ends the process in the flowchart. The response includes information indicating that the request is successfully accepted. For example, in

a case where the request received in step S**1101** is an unlock-with-PIN request, the unlock-with-PIN response includes information indicating that unlocking with the PIN code is successful.

(130) In step S**1160**, the cooperation control unit **503** transmits to the data processing apparatus **101** as a response to the unlock-with-PIN request received in step S**1101**, an unlock-with-PIN response that includes information indicating that unlocking fails because the PIN code does not match, and ends the process in the flowchart.

(131) In step S**1170**, the cooperation control unit **503** transmits to the data processing apparatus **101** as a response to the cooperation state obtaining request received in step S**1101**, a cooperation state obtaining response that includes information about the current cooperation state, and ends the process in the flowchart.

(132) In step S**1180**, the cooperation control unit **503** transmits to the data processing apparatus **101** as a response to the request received in step S**1101**, an error response that includes information indicating that the request is not accepted, and ends the process in the flowchart.

(133) With the process described above, the image forming apparatus **104** can reset the automatic clearing timer when accepting a request from the data processing apparatus **101** without the operation unit **307** accepting an operation. Therefore, for example, the PIN code screen (FIG. **12**E) is not prevented from being displayed, and the user can make the process proceed. The automatic clearing timer is reset only in a case where an accepted request from the data processing apparatus **101** is a request based on an operation on the data processing apparatus **101**, and therefore, the automatic clearing timer is not reset undesirably. For example, a cooperation state obtaining request is a request that is transmitted occasionally without the user operating the data processing apparatus **101**, and therefore, when receiving this request, the image forming apparatus **104** does not reset the automatic clearing timer. The automatic clearing timer of the image forming apparatus **104** may be configured to reset each time one character of a PIN code is input on the data processing apparatus **101**.

(134) With reference to FIG. **13**, a process performed in a case where a user operation is performed on the image forming apparatus **104** will be described. FIG. **13** is a flowchart for explaining the process in the image forming apparatus **104**, and the process is implemented by the CPU **302** of the image forming apparatus **104** loading a program for implementing a control module stored in the ROM **304** or the HDD **305** to the RAM **303** and executing the program. The process in the flowchart illustrated in FIG. **13** is performed asynchronously to the process in the flowchart illustrated in FIGS. **11**A and **11**B.

(135) In step S**1301**, the cooperation control unit **503** detects an event occurring in the image forming apparatus **104**. For example, when the user places an original document on the scanner **313**, a “scanner: an original document placed” event occurs. When the user removes an original document from the scanner **313**, a “scanner: no original document placed” event occurs. When the user presses the placing completed button **1233** in FIG. **12**D, a “UI: the placing completed button pressed” event occurs. When the user presses the cancel button **1232** or the cancel button **1241**, a “UI: the cancel button pressed” event occurs.

(136) In step S**1302**, the cooperation control unit **503** obtains from the event management table **650** information (information defined in the “occurring event” column **652** and in the “changed cooperation state” column **653**) corresponding to the current cooperation state.

(137) In step S**1303**, the cooperation control unit **503** determines whether the event detected in step S**1301** is an event that is to be handled in the current cooperation state. Specifically, the cooperation control unit **503** determines whether the event detected in step S**1301** is included in the “occurring event” column **652** obtained in step S**1302**. If the event is an event to be handled, the cooperation control unit **503** makes the process proceed to step S**1304**; otherwise the cooperation control unit **503** ends the process in the flowchart.

(138) In step S**1304**, the cooperation control unit **503** resets the automatic clearing timer of the image forming apparatus **104**.

(139) In step S**1305**, the cooperation control unit **503** makes the process diverge in accordance with the “changed cooperation state” column **653** obtained in step S**1302**. If the changed cooperation state is “waiting for an original document to be placed”, the cooperation control unit **503** makes the process proceed to step S**1306**. If the changed cooperation state is “PIN code request”, the cooperation control unit **503** makes the process proceed to step S**1310**. If the changed cooperation state is “cancel”, the cooperation control unit **503** makes the process proceed to step S**1320**.
(140) In step S**1306**, the cooperation control unit **503** changes the cooperation state to the “waiting for an original document to be placed” state.
(141) In step S**1307**, the UI control unit **502** displays the original-document placing screen on the operation unit **307** of the image forming apparatus **104** and ends the process in the flowchart.
(142) In step S**1310**, the cooperation control unit **503** changes the cooperation state to the “PIN code request” state. In step S**1311**, the cooperation control unit **503** generates a PIN code. In step S**1312**, the UI control unit **502** displays the PIN code screen on the operation unit **307** of the image forming apparatus **104** and ends the process in the flowchart.
(143) In step S**1320**, the cooperation control unit **503** changes the cooperation state to the “cancel” state. In step S**1321**, the cooperation control unit **503** sets a timer for canceling the cooperation state and ends the process in the flowchart. After the time set for the timer has elapsed, the cooperation control unit **503** deletes the cooperation state and the job settings saved in step S**1108**, and the image forming apparatus **104** transitions to the “not in cooperation” state.
(144) With reference to FIG. **14**, a process for automatic clearing control in the image forming apparatus **104** will be described. FIG. **14** is a flowchart for explaining the process in the image forming apparatus **104**, and the process is implemented by the CPU **302** of the image forming apparatus **104** loading a program for implementing a control module stored in the ROM **304** or the HDD **305** to the RAM **303** and executing the program. The process in the flowchart illustrated in FIG. **14** is performed asynchronously to the processes in the flowcharts illustrated in FIGS. **11**A and **11**B and FIG. **13**.
(145) Note that in this embodiment, an example case where the initial screen of the image forming apparatus **104** is displayed in response to the occurrence of automatic clearing in the image forming apparatus **104** will be described. The initial screen can be set by the user, or the home screen (FIG. **12**B) may be set or a specific application screen may be specified. The operation performed in response to the occurrence of automatic clearing is an example, and another example operation may be performed in which in a case where the image forming apparatus **104** has an authentication function, the user is made to log out of the image forming apparatus **104**.
(146) In step S**1401**, the UI control unit **502** detects the image forming apparatus **104** returning from a sleep state. The sleep state is a state in which only limited components are energized to reduce power consumption in a case where the image forming apparatus **104** is not used for a predetermined time. When the user operates the image forming apparatus **104**, the image forming apparatus **104** returns from the sleep state and becomes ready for use. Although the time during which the image forming apparatus **104** is in the sleep state may be longer than the automatic clearing time or equal to the automatic clearing time, in this embodiment, a description will be given under the assumption that the time during which the image forming apparatus **104** is in the sleep state is longer than the automatic clearing time.
(147) In step S**1402**, in response to the return from the sleep state, the UI control unit **502** displays the initial screen on the operation unit **307** of the image forming apparatus **104**. In step S**1403**, the UI control unit **502** starts counting down by the automatic clearing timer. Thereafter, the automatic clearing timer decreases the remaining time as time passes.
(148) In step S**1404**, the UI control unit **502** monitors the remaining time of the automatic clearing timer, and makes the process proceed to step S**1405** when detecting the remaining time being equal to zero or keeps monitoring the remaining time if the remaining time is more than zero.
(149) In step S**1405**, the cooperation control unit **503** determines whether the image forming

apparatus **104** is in cooperation with the data processing apparatus **101** currently. If the image forming apparatus **104** is in cooperation with the data processing apparatus **101**, the cooperation control unit **503** makes the process proceed to step S**1406**; otherwise the cooperation control unit **503** makes the process proceed to step S**1407**.
(150) In step S**1406**, the cooperation control unit **503** deletes the cooperation state and the job settings saved in step S**1108** and cancels cooperation with the data processing apparatus **101**.
(151) In step S**1407**, the UI control unit **502** displays the initial screen on the operation unit **307** of the image forming apparatus **104** and ends the process in the flowchart.
(152) With the procedure described in this embodiment, in a case where a user operation is performed on either the data processing apparatus **101** or the image forming apparatus **104** while the data processing apparatus **101** and the image forming apparatus **104** are operating in cooperation with each other, the automatic clearing timers of both apparatuses are reset. Accordingly, the automatic clearing function appropriately works while the data processing apparatus **101** and the image forming apparatus **104** are operating in cooperation with each other, resulting in increased usability.
Second Embodiment
(153) In the first embodiment, in a case where automatic clearing occurs in one of the data processing apparatus **101** or the image forming apparatus **104**, automatic clearing is made to occur in the other apparatus in accordance with the automatic clearing time of the other apparatus. In a second embodiment, an example configuration in which in a case where automatic clearing occurs in one of the apparatuses, automatic clearing is made to occur also in the other apparatus will be described. The basic configuration of the second embodiment is the same as that of the first embodiment, and therefore, only differences will be described.
(154) With reference to FIG. **15**, a process for automatic clearing control performed by the data processing apparatus **101** in the second embodiment will be described. FIG. **15** is a flowchart obtained by modifying the flowchart in FIG. **10**. Only differences from FIG. **10** will be described.
(155) In step S**1501**, the MFP control unit **403** determines whether the data processing apparatus **101** is in cooperation with the image forming apparatus **104**. Although determination may be performed in any manner, for example, the cooperation state obtaining response received in step S**708** may be retained, and the cooperation state may be used to perform determination.
(156) Alternatively, determination may be performed on the basis of whether the screen currently displayed on the display **214** is a screen that is displayed during cooperation. If the data processing apparatus **101** is in cooperation with the image forming apparatus **104**, the MFP control unit **403** makes the process proceed to step S**1502**; otherwise the MFP control unit **403** makes the process proceed to step S**1004**.
(157) In step S**1502**, the MFP control unit **403** transmits to the image forming apparatus **104** a notification of the occurrence of automatic clearing.
(158) In step S**1511**, the MFP control unit **403** determines whether a notification of the occurrence of automatic clearing is received from the image forming apparatus **104**. The MFP control unit **403** makes the process proceed to step S**1512** if the notification is received, or makes the process return to step S**1003** if the notification is not received.
(159) In step S**1512**, the MFP control unit **403** determines whether the notification received in step S**1511** is a notification from the image forming apparatus **104** that is in cooperation with the data processing apparatus **101**. If the source of the notification is the image forming apparatus **104**, the MFP control unit **403** makes the process proceed to step S**1004**; otherwise the MFP control unit **403** makes the process return to step S**1003**.
(160) With reference to FIG. **16**, a process for automatic clearing control performed by the image forming apparatus **104** in the second embodiment will be described. FIG. **16** is a flowchart obtained by modifying the flowchart in FIG. **14**. Only differences from FIG. **14** will be described.
(161) In step S**1601**, the cooperation control unit **503** transmits to the data processing apparatus

101 a notification of the occurrence of automatic clearing.
(162) In step S**1611**, the cooperation control unit **503** determines whether a notification of the occurrence of automatic clearing is received from the data processing apparatus **101**. The cooperation control unit **503** makes the process proceed to step S**1612** if the notification is received, or makes the process return to step S**1404** if the notification is not received.
(163) In step S**1612**, the cooperation control unit **503** determines whether the notification received in step S**1611** is a notification from the data processing apparatus **101** that is in cooperation with the image forming apparatus **104**. If the source of the notification is the data processing apparatus **101**, the cooperation control unit **503** makes the process proceed to step S**1407**; otherwise the cooperation control unit **503** makes the process return to step S**1404**.
(164) As described above, according to the second embodiment, in a case where automatic clearing occurs in one of the apparatuses, a notification of the automatic clearing is transmitted to the other apparatus, and this can make automatic clearing occur in both apparatuses that are in cooperation, resulting in increased usability.
Other Embodiments
(165) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)), a flash memory device, a memory card, and the like.
(166) While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but is defined by the scope of the following claims.

Claims

1. An information processing apparatus configured to communicate with a terminal, the information processing apparatus comprising: at least one processor and at least a memory coupled to the at least one processor and having instructions stored thereon, the at least one processor, by executing the instructions, acting as: a control unit configured to execute automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein the control unit executes the automatic clearing for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the operation unit of the information processing apparatus nor on the operation unit of the terminal, and the control unit does not execute the automatic clearing for the information processing apparatus in a case where there is a user operation on the operation unit of the terminal

even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.
2. The information processing apparatus according to claim 1, the at least one processor further acting as: a counting unit configured to count time for which no operation is received from a user, wherein the control unit executes the automatic clearing in a case where the time counted by the counting unit reaches the predetermined time, the counting unit resets counting in a case where a user operation on the operation unit of the information processing apparatus is received, and the counting unit resets counting in a case where a user operation on the operation unit of the terminal is received.
3. The information processing apparatus according to claim 1, wherein the control unit executes the automatic clearing on a basis of an automatic clearing timer, and wherein the automatic clearing timer is based on a user operation on the operation unit of the information processing apparatus and a user operation on the operation unit of the terminal.
4. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of displaying an initial screen.
5. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of clearing a value of a setting set on a displayed screen.
6. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of making a logged-in user of the information processing apparatus log out of the information processing apparatus.
7. The information processing apparatus according to claim 1, wherein the information processing apparatus is an image forming apparatus having at least one of a print function, a scan function, or a copy function.
8. The information processing apparatus according to claim 1, wherein the user operation on the operation unit of the terminal is a user operation received by an application that cooperates with the information processing apparatus and runs on the terminal.
9. The information processing apparatus according to claim 1, wherein the user operation on the operation unit of the terminal is an operation of selecting a button displayed by an application that cooperates with the information processing apparatus and runs on the terminal.
10. The information processing apparatus according to claim 9, wherein the application is an application that gives a print instruction to the information processing apparatus.
11. A method for an information processing apparatus configured to communicate with a terminal, the method comprising: executing automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein the automatic clearing is executed for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the operation unit of the information processing apparatus nor on the operation unit of the terminal, and the automatic clearing is not executed for the information processing apparatus in a case where there is a user operation on the operation unit of the terminal even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.
12. A non-transitory computer-readable storage meduim storing executable instructions, which when executed by one or more processors of an information processing apparatus, cause the information processing apparatus to perform operations comprising: executing automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein the automatic clearing is executed for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the

operation unit of the information processing apparatus nor on the operation unit of the terminal, and the automatic clearing is not executed for the information processing apparatus in a case where there is a user operation on the operation unit of the terminal even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.